

Supplementary material for: Information loss of fixed linear health indicators as a control problem: simplex geometry and reachability limits

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Abstract

Section S1 gives the protocol behind every row of Table 5 of the main text, with the parameters of the code that produced it. Sections S2 to S4 hold material moved out of the main text: two proofs, the two candidate certificates for Proposition 5.1 and what each would need, and the numerical verification of the identities and of the exceptional family of Remark 4.6. Equation, section and proposition numbers without an S refer to the main text.

S1 Protocols for the public-fleet tests

Every number in Table 5 is recomputed by `verify_numbers.py` in the repository named in the main text, and the parameters below are the ones that script passes; where a test was also run at other settings to check stability, both are given. All four protocols read an empirical informative direction as a time derivative of the monitoring channels along normalised life, and so rely on the time-shift population of Section 2, under which that derivative is the sensitivity of Eq. (6).

S1.1 The rotation test on the Severson cells

Channels are capacity at discharge, internal resistance and mean cell temperature. A cell is used if its life is recorded, it has at least 400 cycles, and all three channels are finite throughout; 129 cells qualify. Life is normalised per cell to $\tau \in [0, 1]$ by cycle index. Each channel is differentiated with a Savitzky–Golay filter of polynomial order 3 and window $\min(301, 2\lfloor n/4 \rfloor + 1)$ cycles for a cell of n cycles, and the derivative is divided by the standard deviation of that channel’s residual about the same filter, which whitens the channel by its own measured noise. The whitened derivative is interpolated onto 40 points of $\tau \in [0.15, 0.90]$. One sign per channel, the median sign over every cell and grid point, orients all cells alike; each cell’s vector is then normalised at every grid point, and the fleet mean, renormalised, is the empirical informative direction. Its arc is the sum of the angles between consecutive grid points, and its end-to-end angle is the angle between the first and the last. Positivity is the fraction of grid points at which every component of the fleet mean is positive; it is one at the window used.

S1.2 Why three measured fleets cannot test actuation

Testing actuation needs the operating point to change within a unit while everything else is held. In XJTU-SY each of three operating conditions is applied to its own five bearings, and in PRONOSTIA each of three conditions to its own bearings, so a difference between conditions cannot be told apart from a difference between groups of units. The NASA battery fleet has four cells. These fleets are therefore listed with the reason and no statistic.

S1.3 The regime contrast on C-MAPSS

The training files of FD004 and FD002 are used. Life is normalised per engine by its final cycle. Operating regimes are labelled by standardising the three operating settings over the file and rounding to one decimal; each file has 6 regimes. A sensor is kept if its standard deviation within a (engine, regime) group has a positive median, and each kept sensor is standardised within each regime, so that a regime's level does not enter the direction.

Within one engine, each regime's cycles are resampled to a common count m , the smallest over that engine's regimes, and an engine enters only if $m \geq 2(w + 3)$. Each resampled series is split into its even and its odd cycles, two half-samples of the same regime. On each half the direction is estimated as for the Severson cells, with polynomial order 2 and window w , and interpolated onto 10 points of $\tau \in [0.30, 0.85]$. The angle between two directions is the mean over grid points of the pointwise angle. W is the mean, over the engine's regimes, of the angle between the two halves of the same regime, and B the mean, over pairs of different regimes, of the angle between a half of one and a half of the other. $B - W$ is positive when regimes differ by more than half-samples of one regime differ from each other, which is what an operating point that moves the observed direction predicts. It does not say how the direction is moved. C-MAPSS imposes its flow and efficiency losses as functions of time and lets the regime act only on the thermodynamic response of the sensors [1], so a positive contrast there reads the observation map, not the stress factor of Eq. (2), and cannot test actuation. The table reports the mean of $B - W$ over engines, and the number of engines for which it is positive, at $w = 7$; at $w = 9$ and $w = 11$ every value keeps its sign and exceeds the largest control bias.

The negative control uses the same machinery on fake regimes. Within one regime of one engine with at least $4(w + 3)$ cycles, the four interleaved subsequences of every fourth cycle stand in for four regimes; W compares the first with the second and B the first with the fourth. No contrast exists by construction. Over all three windows and both files the mean of $B - W$ stays within 0.5° of zero, and a paired t test of B against W does not reject equality at any window, the smallest p being 0.35.

S1.4 The separability test on N-CMAPSS, jointly with the channel pairing

The development sets of DS02, units 2, 5, 10, 16, 18 and 20, and of DS01, units 1 to 6, are used. The operating point is held fixed by clustering flight points, by k -means with 6 clusters, on standardised altitude, Mach number and throttle-resolver angle; life is divided into 2 stages of equal width in cycles. A (stage, cluster) cell enters only with at least 400 points.

Within a cell each sensor is regressed on standardised remaining life, and the slope is divided by the standard deviation of that sensor's residual, so that the vector of slopes is the informative direction at a fixed operating point rather than a residual after standardisation. The vector is normalised, and the logarithms of the absolute values of its components, centred on their mean,

Table S1: The parameters of each test as run.

test	parameters
Severson rotation	129 cells with at least 400 cycles; Savitzky–Golay order 3, window 301 cycles or $2\lfloor n/4 \rfloor + 1$ if smaller; 40 points on $\tau \in [0.15, 0.90]$
C-MAPSS regime contrast	FD004 and FD002 training sets; 6 regimes each; order 2, window $w = 7$, stability at $w = 9, 11$; 10 points on $\tau \in [0.30, 0.85]$; engine enters if $m \geq 2(w + 3)$; control needs $4(w + 3)$ cycles in one regime
N-CMAPSS separability with the pairing	DS02 units 2, 5, 10, 16, 18, 20 and DS01 units 1–6; 6 clusters, 2 stages, at least 400 points per cell; sweep over seven (cluster, stage) settings with at least 300 points

are what the test compares. The logarithm is the point: under the separable form of Lemma 4.4, read through the pairing of channels with mechanisms in Eq. (5), each component is a product of an operating-point factor and a state factor, so in logarithms the two act additively, and the absence of interaction is what the two jointly predict. A sensor map that mixes mechanisms with weights depending on the operating point breaks that additivity as surely as a non-separable stress factor does. In N-CMAPSS the operating history enters the damage through the onset of abnormal degradation [2], and the health parameters the files carry, which are the damage the simulator imposed, are constant within every one of the 446 flights of DS02 and 553 of DS01. The damage therefore does not respond to the operating point within a flight, and the sensor map is the only source of any dependence on the operating point that the test finds.

Each cell is estimated twice, from its even and from its odd points. The mean of the two is the signal; their half-difference measures noise, and applying the same decomposition to it gives each effect its own noise floor. The decomposition is the ordinary two-way one over the cells, into a grand mean, a stage effect, an operating-point effect and an interaction, with sums of squares taken over cells and sensors and pooled over units. With 6 clusters and 2 stages the operating-point effect and the interaction each have 5 degrees of freedom per sensor and the stage effect 1, so the reported ratio of interaction to operating-point effect compares quantities with equal degrees of freedom. That ratio is 1.65 for DS02 and 1.84 for DS01. Against its own noise floor the interaction is 566 times larger on DS02 and 110 times on DS01, so it is not noise. The test was repeated at 3, 4, 6, 8 and 10 clusters with 2 stages, at 6 clusters with 3 stages and at 4 clusters with 4 stages, with at least 300 points per cell; the interaction exceeds the operating-point effect in every configuration, by a factor of at least 1.42.

S2 Proofs of Propositions 2.3 and 3.3

Proof of Proposition 2.3. In whitened coordinates the linearised record is $\tilde{y}_k = \mathbf{d}_k R + \epsilon_k$ with $\epsilon_k \sim N(0, I_d)$ independent, so the Fisher information it carries about R is $\sum_k \mathbf{d}_k^\top \mathbf{d}_k = \sum_k \|\mathbf{d}_k\|^2$ and an oracle reading the whole record obtains $\frac{1}{2} \log(1 + G)$ nats. The fixed indicator returns the scalars $z_k = \mathbf{v}^\top \tilde{y}_k = (\mathbf{v}^\top \mathbf{d}_k) R + \mathbf{v}^\top \epsilon_k$, whose noise terms $\mathbf{v}^\top \epsilon_k$ are standard normal and independent because $\|\mathbf{v}\| = 1$, so the compressed record carries Fisher information $\sum_k (\mathbf{v}^\top \mathbf{d}_k)^2$. Since $\mathbf{v}^\top \mathbf{d}_k = \|\mathbf{d}_k\| \cos \psi_k$,

$$\text{Var}[R] \sum_k (\mathbf{v}^\top \mathbf{d}_k)^2 = \text{Var}[R] \sum_k \|\mathbf{d}_k\|^2 \cos^2 \psi_k = G(1 - S),$$

and subtracting the two mutual informations gives $\frac{1}{2} \log(1 + G) - \frac{1}{2} \log(1 + G(1 - S)) = -\frac{1}{2} \log \frac{1+G-GS}{1+G} = -\frac{1}{2} \log(1 - WS)$. $\square$

Proof of Proposition 3.3. (i) As $\gamma \rightarrow \infty$ we have $w \rightarrow 1$, and (13) gives

$$\ell \rightarrow -\frac{1}{2} \log \cos^2 \psi = -\log \sum_i \sqrt{p_i q_i} = D_B$$

by Proposition 3.2. The convergence is uniform, which is what carries the minimisers with it. Positivity keeps p_τ interior, and $\sqrt{q_i} \geq q_i$ with $\sum_i q_i = 1$ gives $\cos \psi = \sum_i \sqrt{p_i q_i} \geq \min_i \sqrt{p_i}$ for every q in the simplex, so $\cos \psi$ is bounded away from zero uniformly in q and in τ and D_B stays finite. Both objectives are continuous on a compact set and uniformly close on it, so their minimisers converge. (ii) By (14), $\bar{\ell} = \frac{1}{2} \bar{w} \mathbb{E}_\mu[\psi^2] + O(\psi^4)$ with $d\mu = w d\tau / \bar{w}$; minimising a μ -weighted mean square of a geodesic distance is the definition of the weighted Fréchet mean, and $\psi = \frac{1}{2} d_{\text{FR}}$ converts $\frac{1}{2} \bar{w} \mathbb{E}_\mu[\psi^2]$ into $\frac{1}{8} \bar{w} \mathbb{E}_\mu[d_{\text{FR}}^2]$. $\square$

S3 Two candidate certificates for Proposition 5.1

Continuous differentiability is a convenience of the proof in the main text, not a requirement of the argument. A continuous viscosity subsolution of the Hamilton–Jacobi–Bellman equation of the Dinkelbach problem that is non-positive on the condemnation surface lies below the value function by the comparison principle, because the drift points out of the state region across that surface, and a Lipschitz V satisfying (27) with Clarke’s generalised gradient in place of ∇V certifies in the same way [3]. The obstacle is therefore not smoothness but exhibiting a candidate whose inequality can be checked.

We tried two. The value function of the Dinkelbach problem of Section 5 satisfies (27) with equality in the viscosity sense [3], so it is the natural candidate, but only a grid approximation of it is available. Computed on a 41^3 grid and tested by finite differences on a 61^3 one, (27) fails at about 45% of test points by up to 2.9×10^{-3} . Halving λ moves that violation by 0.6%, so it does not come from asking too much of the bound. It is the error of a first-order scheme on a coarse grid, differentiated on a finer one, and Section 6.6 shows how large that error still is on grids of this size. A certificate from this route needs the discrete value function shifted by a certified bound on its own residual, or an error estimate for the monotone scheme that produced it. At the opposite extreme an affine $V(x) = c_0 - a_0^\top x$ has an exactly constant gradient and makes every requirement linear, so the best certifiable λ is a linear program; over 167,936 constraints it returns $V \equiv 0$ and only the pointwise bound $\min_{x,u} \ell$, which is vacuous here because the reachable floor falls to within sampling resolution of zero at mid-life. Deleting the two boundary requirements lets that program run to any cap imposed on it, which identifies them as what binds: trajectories of (2) increase in every coordinate, so an affine V is monotone along all of them at once and $V(x_0) \geq 0 \geq V|_{\text{failure}}$ forces it to decrease, while the near-aligned states, where the running cost vanishes, demand a local increase.

The two failures are complementary, and together they say what a certificate needs. The value function varies its gradient as the two demands require, but its computable approximations carry an error of the size of the effect; affine functions are exact but cannot vary their gradient at all. A certificate therefore needs a gradient that varies and an error that is controlled, which points at a discrete value function with a certified residual bound, or at a smooth under-approximation of the value function obtained by sum-of-squares programming when the plant data are polynomial.

S4 Numerical verification

S4.1 The identities

The analytical results of the main text were checked numerically before they were used. The rotation identity (20) holds to relative error below 1.1×10^{-7} over the five input–channel pairs $(d, m) = (2, 1), (3, 2), (3, 1), (4, 2)$ and $(5, 3)$; the replicator form to 6.9×10^{-8} ; Theorem 3.6 to 5.1×10^{-6} , limited by finite differencing; and (17) to 1.1×10^{-16} .

S4.2 The exceptional family of Remark 4.6

Integrating the construction of Remark 4.6 for $d = 2$ to 8 with one input holds the misalignment below 1.5×10^{-6} degrees and the mean loss below 10^{-15} . These families carry no prior variance of their own, so their losses are evaluated at a fixed $w = 0.9$ rather than at a γ the plant supplies; the bound has fifteen orders of magnitude of headroom and does not turn on the choice. That is against 1.2×10^{-5} to 5.4×10^{-4} for the best constant input on the same plant.

The family is a knife edge. Perturbing the shape coefficients ν by 1% raises the mean loss to 3.2×10^{-4} under the myopic policy, against 1.8×10^{-4} for the best constant input on the same perturbed plant. At one percent of shape error, steering is already worse than not steering. Closing the loop halves the damage but does not repair it, so the fragility belongs to the construction and not to the open-loop schedule. Mistuning the activation energies instead is tolerated to about two percent.

References

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